

MUHAMMAD SULEIMAN QURESHI

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EDUCATION

B.S. Electrical Engineering
Minor: Computer Science
Habib University, Karachi

Aug 2020 – Dec 2024
CGPA: 3.88/4.00

FINAL YEAR PROJECT

Autonomous EV Tow Truck Control System

Industry Collaboration: IMC Toyota

- Retrofitted an electric tow truck with electronic sensing and actuation by designing custom PCBs, power electronics, and embedded control hardware
- Performed system identification to obtain accurate vehicle dynamics models
- Developed and tuned PID controllers for steering and velocity regulation
- Implemented Model Predictive Path Integral (MPPI) control using ROS2 NAV2 for real-world navigation
- Designed a hierarchical PID-MPPI control architecture: inner-loop PID for real-time steering/velocity regulation, outer-loop MPPI for trajectory planning and obstacle avoidance
- Achieved steering accuracy of $\pm 1^\circ$ and velocity error of $\pm 0.3\text{m/s}$ under real-world constraints

PROJECTS

Visual Fault Detection for Plastic Parts (Dawlance)

Developing a computer vision system to automatically detect and classify visual defects in injection-molded components, projected to reduce quality-related losses by PKR 7.2 million annually. Using a PatchCore pipeline with WideResNet-101 at 1024×1024 resolution, trained on 640 nominal images only (no defect data), the system achieved 100% Recall (0/250 missed defects) and $F1 = 0.96$ on the industrial test set. Published at CVC 2026.

6-DOF Industrial Robot Arm (Toyota & Dawlance)

Developing a six-degree-of-freedom robotic manipulator focusing on kinematic modeling and control-oriented system (0.04mm tolerance) integration for industrial use.

Visual Vehicle Repair Management System (Toyota)

Developing a machine learning-enabled computer vision system for automated vehicle damage documentation and repair process digitization.

Vision-Guided Pick & Place Robotic Pipeline

Implemented a vision-guided pick-and-place pipeline integrating camera-based object localization with robotic manipulation integrating hardware with MATLAB.

Two-Wheeled Self-Balancing Robot

Implemented and tuned real-time PID control on embedded hardware to stabilize an inverted pendulum system.

RESEARCH EXPERIENCE

Symbolic Health Equations for Unsupervised
Gearbox Fault Detection via Kolmogorov-Arnold Autoencoders

(Under Review at [IEEE ICIEA 2026](#))

MPPI Path Following with Foothold-Quality-Aware Cost
for Quadruped Navigation in Unstructured Terrain

(Under Review at [AIM 2026](#))

Deployment-Oriented Memory-Based Visual Inspection
of Injection-Molded Parts

(Accepted at [CVC 2026](#))

Development of a DC-DC converter based nanogrid platform

Presented in DURS 2023

**Effects of Parasitic Elements in High Frequency GaN-based
DC-DC Converters for Electric Vehicle Applications [PDF]**

Presented in INMIC 2023

Summer Tehqiq Researcher

2022–2024

Multi-Year funded research summers across automation, robotics, power electronics, and electric vehicles

AWARDS

Dean's List (4x)

AP Scholar with Distinction

PROFESSIONAL EXPERIENCE

Assistant Manager

February 2025 - Present

Takhleeq Office, Habib University

- **R&D & Emerging Tech:** Led research in advanced engineering, materials, and automation; developed prototypes including an autonomous tow truck, a robotic arm, and industrial automation solutions.
- **Project Leadership:** Managing three concurrent projects-Aesthetic Fault Identification (Dawlance), Industrial Robot Arm Development (Dawlance & Toyota), and Vehicle Repair Management System (Toyota)-while ensuring technical feasibility, timely delivery, budget adherence, and systematic validation and optimization.
- **Collaboration & Delivery:** Co-led industry-academia initiatives, secured funding through proposal writing, and aligned cross-functional teams to deliver scalable solutions with structured documentation, risk management, and PMO support.

Researcher

January 2025 - February 2025

Habib University

- Research and development of an Autonomous Tow Truck as a continuation of my final year project.

Software Engineering Intern

June 2023 - September 2023

Motive (formerly known as KeepTruckin)

- Enhanced vehicle simulation scripts to handle multiple ECUs, improving functionality and testing reliability.
- Collaborated closely with a cross-functional team to integrate the new feature within the existing architecture.

Teaching Assistant for Data Structures and Algorithms (*Spring 2022 & 2023*),
Programming Fundamentals (*Fall 2022*), Signals & Systems (*Spring 2023*),
and Digital Logic & Design (*Spring 2024*)

January 2021 - April 2024

Habib University

- Assisted over 100 students across 4 courses
- Facilitated students in their labs with their queries
- Set up Teaching Hours outside of lab timings for students

TECHNICAL SKILLS

Programming: Python, C++, MATLAB, Bash

Robotics & Control: ROS2, NAV2, PID, MPC, MPPI, EtherCAT/SOEM

AI / ML: Fault Detection, Vibration-Based Condition Monitoring, Anomaly Detection, PyTorch, TensorFlow, CNNs

Simulation: Gazebo, Simulink, LabVIEW

Embedded Systems: Arduino, FPGA, TivaC, Raspberry Pi

EXTRACURRICULAR

Lead of IEEE Student Chapter

March 2023 - March 2024

Habib University

- Spearheaded initiatives to promote engagement in technology through events, workshops, and seminars.

Orientation Leader 2021

July 2021 - August 2021

Habib University

- Coordinated New Student Orientation 2025, mentoring incoming students and leading group activities to foster a welcoming environment.